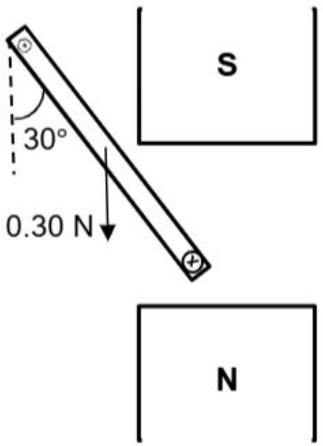
4 A rigid square coil of sides 50 mm has 100 turns. It hangs vertically such that it can rotate freely about one side.

When a current of 0.40 A passes through the coil, it hangs in equilibrium at an angle of 30° to the vertical with its lowest side in the vertical magnetic field of a magnet. The weight of the coil is 0.30 N.



What is the flux density of the field between the magnets?

	A	0.038 T	**B**	0.043 T	**C**	0.075 T	**D**	0.087 T

L2 **Answer: B**

Magnetic force on the side of the coil in the magnetic field **acts horizontal to the right**.

By taking moments at the pivot,
Total clockwise moment = Total anticlockwise moment
Weight of coil x perpendicular distance = magnetic force x perpendicular distance
$Mg \times (L/2 \sin 30°) = (NBIL) \times (L \cos 30°)$
$0.30 \times [\, 25 \times 10^{-3} \sin 30° \,] = [100 \times B \times 0.40 \times 50 \times 10^{-3}\,] \times [\, 50 \times 10^{-3} \times \cos 30° \,]$
$B = 0.043$ T